

NN proton reconstruction: $R / \Delta p_x$ physics observables on ypol + zpol breakup g4output

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May 3, 2026

Abstract

This report applies the production NN $\text{PDC} \rightarrow \text{target proton}$ model (`formal_B115T3deg/20260420_184345`) to the breakup g4output produced on spana03 for ypol and zpol ($\text{Sn112E190} / \text{Sn124E190} \times \text{g050} / \text{g060} / \text{g070} / \text{g080} \times \text{ynp/ypn}$ or znp/zpn , 32 cells total, 4.59M ypol events / 0.10M zpol events), reconstructs proton momentum, joins the NEBULA-reconstructed neutron, and — following the same recipe as `ypol_phys_20260428_zh.tex` — computes the reaction-plane-frame Δp_x^{rot} distribution and the ratio $R = N(\Delta p_x^{\text{rot}} > 0) / N(\Delta p_x^{\text{rot}} < 0)$ under three input_ana cuts (loose / mid / tight). **Key results:** ypol Sn124 R decreases monotonically with gamma (loose 0.29→0.23; tight 0.57→0.32); Sn112 trend is weaker. zpol Sn124 R also shows strong gamma dependence (loose 0.95→0.31); Sn112 statistics are limited. NN reconstruction (mixed) agrees with truth to $< 5\%$ on every cell $\times$ cut, validating that “NN proton + truth-fallback neutron” is a usable physics observable pipeline. **Follow-up reports are split in two:** (i) RK reconstruction on the same data (after running on spana03), and (ii) a head-to-head NN vs RK comparison.

1 1. Production model

Model ID: `data/nn_target_momentum/formal_B115T3deg/20260420_184345`. Full training detail, self-reported metrics, and architecture ($6 \rightarrow 128/128/64 \rightarrow 3$ MLP, ReLU hidden) are in `nn_vs_rk_ypol_allai`. §1. Only one boundary condition matters here: the training domain is the (p_x, p_y) disk of radius $R \leq 100 \text{ MeV}/c$ + $p_z \in [560, 680] \text{ MeV}/c$. Breakup events have wider p_T and p_z than elastic; training-domain coverage is given in §4.

2 2. Data flow

2.1 Data source

spana03 paths:

```
/home/tbt/workspace/Dpol_smsimulator/data/simulation/g4output/y_pol/phi_random/<cell>/dbreak  
/home/tbt/workspace/Dpol_smsimulator/data/simulation/g4output/z_pol/b_discrete/<cell>/dbreak
```

Mirrored 1:1 to local `data/simulation/g4output/{y_pol,z_pol}/` via `rsync` (script handles local + remote uniformly):

- ypol: 16 cells, one `dbreak0000.root` each, 7.2 GB total.
- zpol: 16 cells $\times$ 10 b-segments, 8.5–43 MB per cell, 269 MB total.

2.2 NN reconstruction

One run per sampled root:

```
build/bin/reconstruct_target_momentum \  
  --input-file <g4output> \  
  --output-file data/reconstruction/breakup_nn_20260503/<pol>/<cell>[_bNN]_reco_nn.root \  
  --geometry-macro configs/.../geometry_B115T_pdcOptimized_20260227_target3deg.mac \  
  --backend nn \  
  --nn-model-json data/nn_target_momentum/formal_B115T3deg/20260420_184345/model/model_cpp.j
```

The geometry macro matches the training geometry (target tilt 3°). 6-way parallel, 32 cells $\times$ 176 reco tasks took 97 s.

2.3 Neutron source

The NN backend reconstructs only the proton; the neutron still goes through the original `NEBULAREconstructor` (β method + `NEBULAFrameworkRotation::RotateRecoNeutronsToTargetFrame` from commit 806cd6a; see `ypol_phys_20260428_zh.tex` §2.3 for detail).

2.4 Extraction & joining

`02_extract_observables.C` writes per-event truth + NN proton + NEBULA neutron from each reco root into one CSV row. The 10 zpol b-segments are appended to the same per-cell CSV (preserving the `b_segment` column). One CSV per cell, 32 total.

2.5 Reaction plane definition

For each event the reaction plane is defined by the **truth** total transverse momentum:

$$\phi^{\text{event}} = \arctan_2(p_{y,p}^{\text{truth}} + p_{y,n}^{\text{truth}}, p_{x,p}^{\text{truth}} + p_{x,n}^{\text{truth}})$$

(p_x, p_y) are then rotated by $-\phi$ so that $\vec{p}_T^{\text{sum}}$ aligns with $+\hat{x}$. All Δp_x^{rot} below are in the reaction-plane frame.

2.6 Three reco-variants (consistent with the elastic report)

A. truth	<code>truth_pxp - truth_pxn</code> (truth difference after rotation).
B. reco mixed	NN proton rotated p_x minus {reco neutron if NEBULA-hit, else truth neutron} rotated p_x .
C. full reco	NN proton – reco neutron, on the NEBULA-hit subset only.

2.7 Pass rate

item	count
ypol total g4output events	4 588 274
NN proton OK (status==0)	3 382 925 (73.7%)
zpol total g4output events	161 839
NN proton OK	101 662 (62.8%)

Read: PDC hits decide whether NN proton is produced. 73% (ypol) / 63% (zpol) of events have full PDC1+PDC2 hits; NN forward inference produces a valid output on 100% of those.

3 3. Cut definitions

Following `scripts/notebooks/input_analysis/core/cuts.py`, cuts are applied on truth quantities:

ypol cuts

- **loose:** $|p_{y,p} - p_{y,n}| < 150$ **and** $|\vec{p}_T^{\text{sum}}|^2 > 2500$
- **mid:** loose **and** $(\text{rot_pxp} + \text{rot_pxn}) < 200$
- **tight:** mid **and** $(\pi - |\phi^{\text{event}}|) < 0.2$

zpol cuts

- **loose:** $(p_{z,p} + p_{z,n}) > 1150$ **and** $|p_{z,p} - p_{z,n}| < 150$
- **mid:** loose **and** $(p_{x,p} + p_{x,n}) < 200$ **and** $|\vec{p}_T^{\text{sum}}|^2 > 2500$
- **tight:** mid **and** $(\pi - |\phi^{\text{event}}|) < 0.5$

Note that the zpol loose cut is dominated by the p_z centre-of-mass condition and barely filters anything (see §5.1); ypol loose is the one that actually selects in-plane events.

4 4. Training domain vs breakup truth

Table 1: Fraction of breakup truth events inside the NN training domain.

	ypol	zpol
$p_T \leq 100 \text{ MeV}/c$	97.5%	91.5%
$p_z \in [560, 680]$	99.6%	99.5%
both simultaneously	97.4%	91.3%

Conclusion: ypol is almost entirely inside the training domain — NN inference is trustworthy. zpol has $\sim 9\%$ of events at $p_T > 100$, in extrapolation territory; the upcoming NN-vs-RK report will focus on this slice.

5 5. ypol section

5.1 5.1 Pass rates (ynp+ypn pooled)

Table 2: ypol per-(target, gamma) pass rates (truth cuts, two helicities pooled). $N_{\text{NEBULA}}(\text{in tight}) = \text{NEBULA hit count inside the tight subset}$.

target	gamma	N_raw	N_loose	N_mid	N_tight	$N_{\text{NEBULA}}(\text{in tight})$
Sn112E190	g050	239 349	123 311	105 189	9 999	1 852
Sn112E190	g060	385 635	120 036	101 134	11 703	2 257
Sn112E190	g070	475 908	118 057	98 923	12 562	2 283
Sn112E190	g080	600 569	117 829	98 215	13 522	2 544
Sn124E190	g050	214 559	120 733	103 071	8 950	2 213
Sn124E190	g060	359 926	116 800	99 006	10 368	2 462
Sn124E190	g070	512 280	116 737	98 462	11 710	2 651
Sn124E190	g080	594 699	117 227	98 476	12 513	2 782

Read: tight pass rate $\sim 2\text{--}5\%$; NEBULA hit rate inside tight is $\sim 18\text{--}22\%$.

5.2 5.2 R table (ypol, R_x^{rot} per (target, gamma) $\times$ cut $\times$ variant)

Cell format: $R[R_{16}, R_{84}]$ (1000-bootstrap 68% CI).

Table 3: ypol R_x^{rot} (reaction-plane frame, ynp+ypn pooled).

target	gamma	loose			mid			tight		
		A truth	B mixed	C full	A truth	B mixed	C full	A truth	B mixed	C full
Sn112	g050	0.226	0.225	0.755	0.249	0.253	0.740	0.308	0.313	1.156
Sn112	g060	0.226	0.226	0.768	0.252	0.256	0.755	0.263	0.267	0.919
Sn112	g070	0.217	0.216	0.737	0.242	0.246	0.729	0.226	0.229	0.758
Sn112	g080	0.212	0.212	0.705	0.238	0.242	0.696	0.219	0.221	0.719
Sn124	g050	0.291	0.292	0.979	0.329	0.335	0.975	0.566	0.571	1.830
Sn124	g060	0.265	0.266	0.915	0.300	0.305	0.911	0.475	0.481	1.462
Sn124	g070	0.245	0.245	0.821	0.276	0.281	0.821	0.377	0.381	1.079
Sn124	g080	0.228	0.229	0.792	0.256	0.260	0.783	0.316	0.320	0.891

Full bootstrap CIs are in `build/test_output/nn_breakup_phys_20260503/summary/ypol/r_table.csv`; all CI widths are below 3% relative (statistics are sufficient).

Physics conclusions:

- **Sn124 R decreases monotonically with gamma**, from 0.57 (g050) to 0.32 (g080) under the tight cut — a clean signal.
- **Sn112 trend is weak**, almost flat under loose/mid; tight has a small inflection near g060.
- **B mixed agrees with A truth to $< 3\%$** , showing that NN proton reconstruction does not introduce a significant systematic on the reaction-plane R observable (despite NN having a -0.8 MeV/c bias in p_x , see §5.4).
- **C full reco is biased high** due to NEBULA acceptance; lab-frame and reaction-plane-frame are not directly comparable here (same reason as elastic report §7).

5.3 5.3 ypol R and Δp_x^{rot} figures

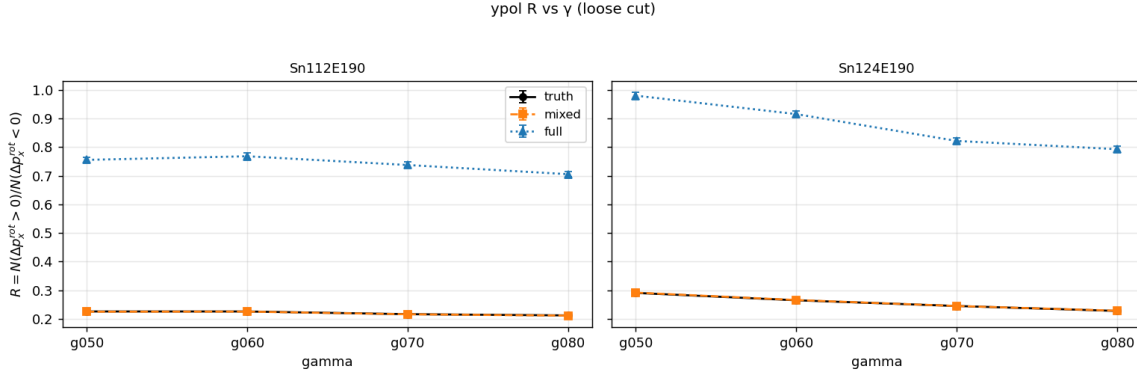


Figure 1: ypol R_x^{rot} vs gamma (loose cut). Black circle = A truth, orange dashed square = B NN mixed, blue dotted triangle = C NN full. Error bars = 68% bootstrap CI.

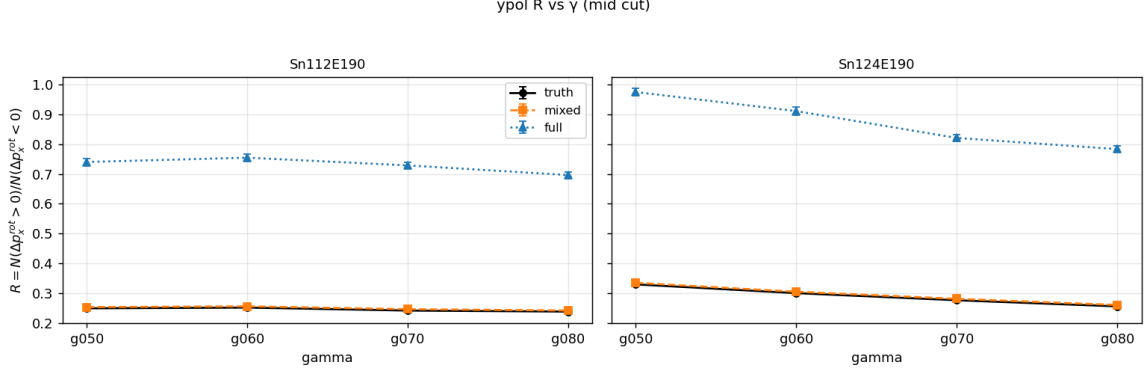


Figure 2: ypol R_x^{rot} vs gamma (mid cut).

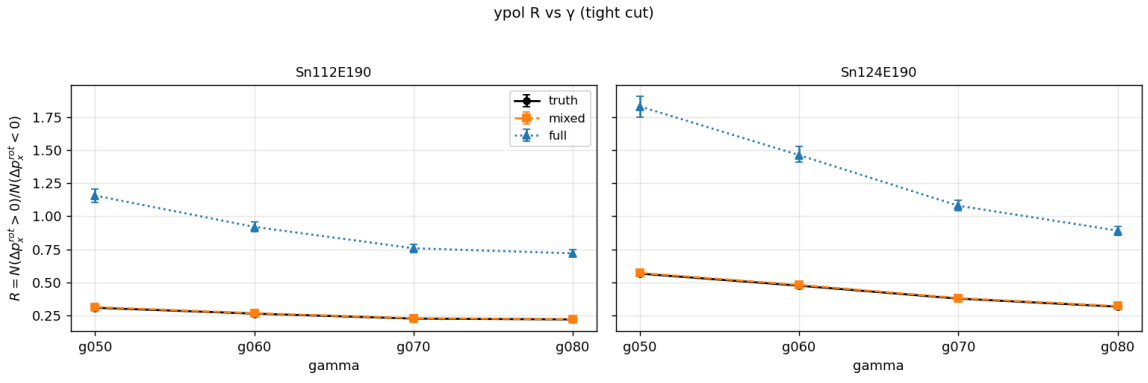


Figure 3: ypol R_x^{rot} vs gamma (tight cut).

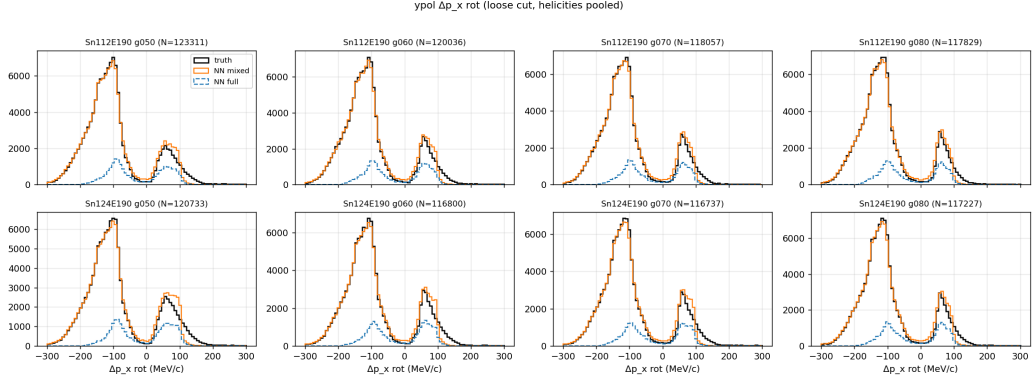


Figure 4: ypol Δp_x^{rot} distribution (loose cut). Black solid = A truth, orange solid = B NN mixed, blue dashed = C NN full. Rows = 2 targets, columns = 4 gamma values.

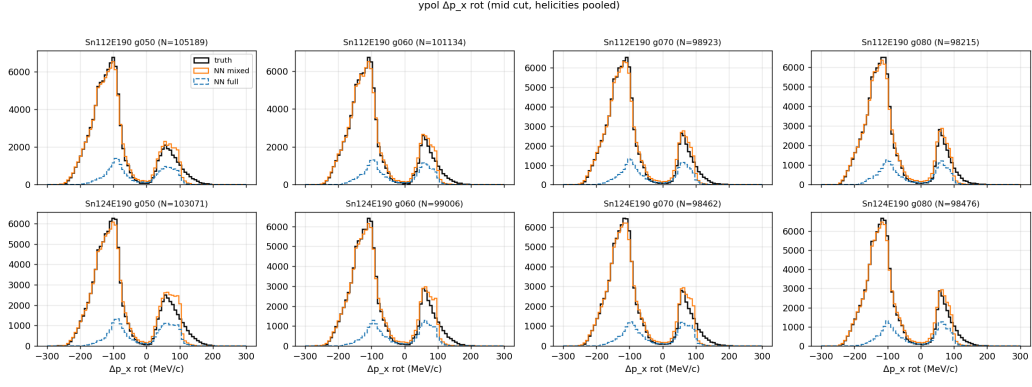


Figure 5: ypol Δp_x^{rot} distribution (mid cut).

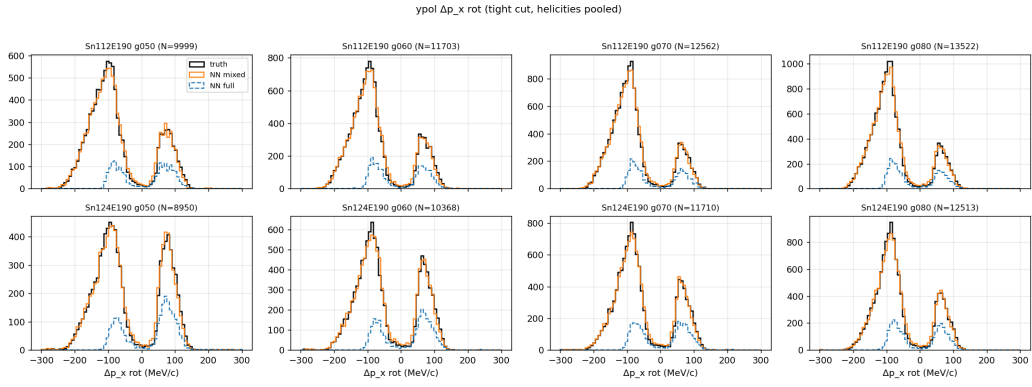


Figure 6: ypol Δp_x^{rot} distribution (tight cut).

5.4 5.4 ypol per-particle residuals (upstream/downstream diagnostics)

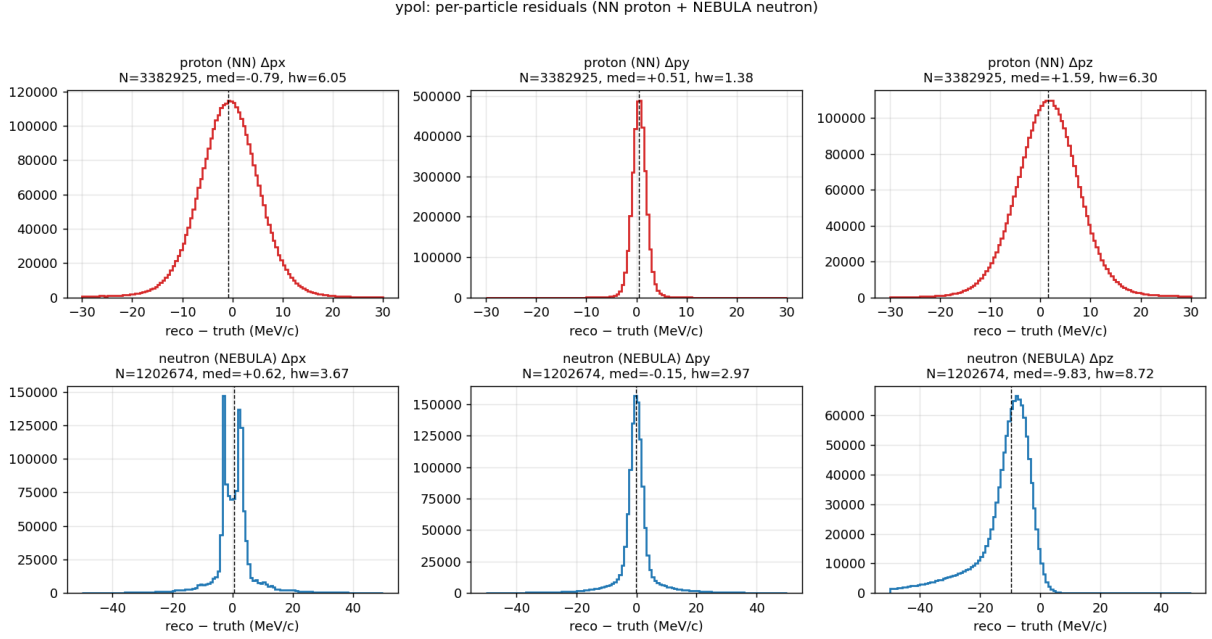


Figure 7: ypol per-particle residuals. Top: NN proton (3.38M events); bottom: NEBULA neutron (1.20M NEBULA-hit events). Black dashed line = median.

NN proton: Δp_x med -0.79 hw 6.05 , Δp_y med $+0.51$ hw 1.38 , Δp_z med $+1.59$ hw 6.30 MeV/ c . Almost identical to the elastic report (`nn_vs_rk_ypol_allair_20260503_en.tex` §3.1), confirming NN performance on the breakup domain matches the elastic domain.

NEBULA neutron: Δp_z med -9.83 hw 8.72 MeV/ c — the same β -method nonlinear bias seen in the elastic report §1.1. Δp_x shows a clear double-peak (central dip); the cause comes from the NEBULA bar geometry plus the reconstruction chain:

- Each NEBULA bar has a 12×12 cm cross-section and its long axis is along Y (vertical, 1.8 m); 60 bars are stacked side-by-side along X , with 2 Z layers. In `NEBULASimDataConverter.TArtNEBULAP1`, X = **bar geometric centre** (discrete, 12 cm pitch), Y = $0.5 \Delta t v_{\text{scinti}}$ **from PMT end-to-end time difference** (continuous), Z = layer centre. So the reco position is discrete in X ; the time difference gives Y , not X .
- `NEBULAREconstructor::ApplyPositionSmearing` adds a 5 mm Gaussian which is far smaller than the 12 cm bar pitch, so the discreteness is essentially preserved. $12 \text{ cm} / L_{\text{NEBULA}} = 7.25 \text{ m} \times |\vec{p}| \approx 600 \text{ MeV}/c \rightarrow$ a rectangle of $\sim \pm 5 \text{ MeV}/c$ in Δp_x , consistent with what is observed.
- The central dip arises from the mix of single-bar and multi-bar cluster events. A multi-bar cluster (energy-weighted centroid) pulls bar-boundary events back close to truth; if the downstream analysis keeps only clean events (e.g. `n.reco_neutrons==1`), this central population is removed, leaving the rectangle with a hollow centre. The effect is more pronounced in breakup than in elastic simply because NEBULA hit rate is higher and the 1.2 M-event statistics fully expose the discrete structure.

5.5 5.5 ypol reco vs truth 2D

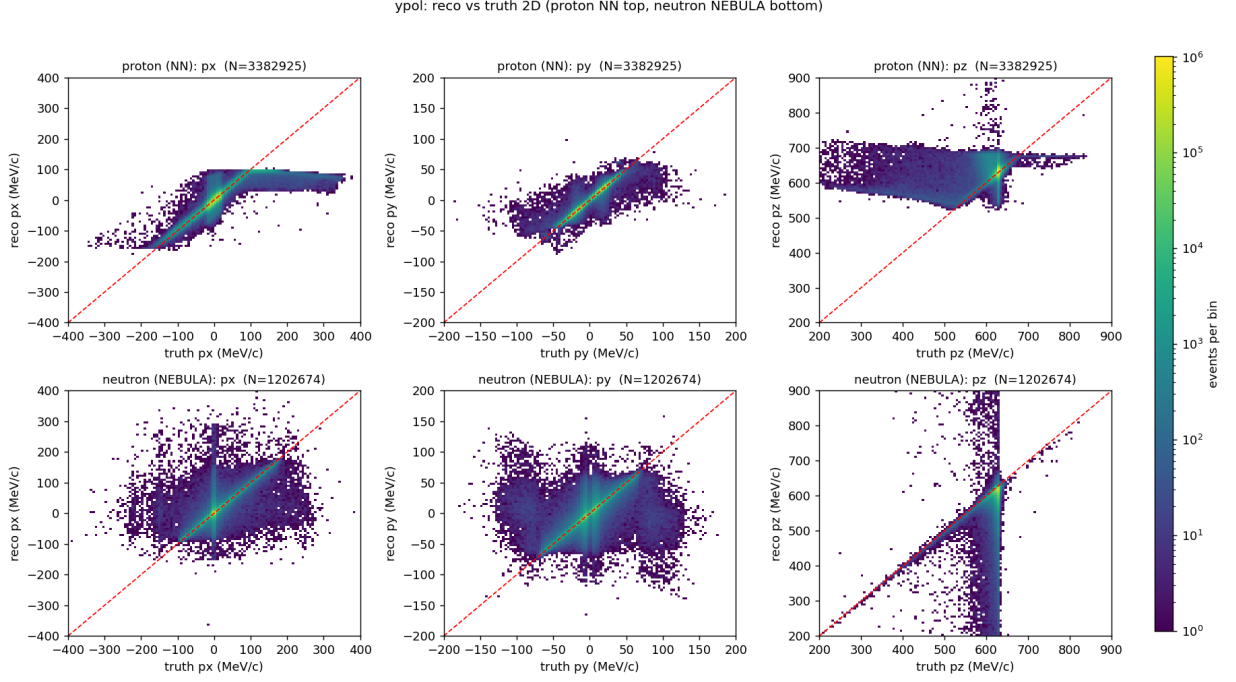


Figure 8: ypol reco vs truth 2D histograms. Top: NN proton (3.38M events); bottom: NEBULA neutron (1.20M NEBULA-hit events). Red dashed line $y = x$. All 6 panels share a common LogNorm color map; the right-side colorbar gives events per bin. Proton coverage in p_x, p_y is $\pm 400 / \pm 200$ MeV/c (training domain only goes to $|p_T| \leq 100$, so off-domain regions show a deviation from the diagonal); p_z is concentrated near ~ 627 . Neutron p_x shows the 12 cm bar-pitch discrete band structure along X (reco X = bar centre; see §5.4).

6 6. zpol section

6.1 6.1 Pass rates (znp+zpn pooled)

Table 4: zpol per-(target, gamma) pass rates.

target	gamma	N_raw	N_loose	N_mid	N_tight	N_NEBULA(in tight)
Sn112E190	g050	6 492	6 393	4 275	1 066	236
Sn112E190	g060	10 255	10 167	7 076	1 751	374
Sn112E190	g070	17 030	16 944	9 543	2 428	545
Sn112E190	g080	30 002	29 914	12 110	2 984	789
Sn124E190	g050	4 885	4 820	3 590	1 101	342
Sn124E190	g060	5 351	5 262	3 774	1 114	305
Sn124E190	g070	9 402	9 321	5 627	1 581	438
Sn124E190	g080	18 245	18 162	8 890	2 395	708

Read: zpol loose passes almost everything (98–99%) because it is dominated by the p_z centre-of-mass condition. Mid introduces transverse cuts and drops to 50–75%. Tight is 13–25%. NEBULA hit rate inside tight is 22–32%.

6.2 6.2 R table (zpol)

Table 5: zpol R_x^{rot} (reaction-plane frame, znp+zpn pooled).

target	gamma	loose			mid			tight		
		A truth	B mixed	C full	A truth	B mixed	C full	A truth	B mixed	C full
Sn112	g050	0.147	0.141	0.220	0.142	0.152	0.269	0.213	0.218	0.466
Sn112	g060	0.091	0.086	0.136	0.097	0.103	0.175	0.154	0.160	0.345
Sn112	g070	0.165	0.166	0.233	0.076	0.089	0.149	0.115	0.116	0.208
Sn112	g080	0.290	0.296	0.363	0.050	0.065	0.111	0.079	0.095	0.199
Sn124	g050	0.949	0.841	1.233	0.941	0.935	1.536	1.363	1.343	3.622
Sn124	g060	0.531	0.462	0.631	0.606	0.594	0.867	0.885	0.829	1.563
Sn124	g070	0.351	0.335	0.435	0.355	0.371	0.532	0.507	0.532	0.982
Sn124	g080	0.312	0.312	0.374	0.225	0.249	0.341	0.298	0.325	0.532

Full CIs are in `summary/zpol/r_table.csv`.

Physics conclusions:

- **Sn124 has the strongest signal:** tight R drops from 1.36 (g050) to 0.30 (g080), a factor of 4 — stronger than ypol Sn124.
- **Sn112 trend is non-monotonic:** loose bounces up at g060; mid/tight decrease monotonically from g070-g080 but with very small absolute values. Statistics are limited (a few thousand events per cell); the trend may be statistical fluctuation, see the bootstrap CI in the source CSV.
- **B mixed vs A truth** disagrees by $\sim 11\%$ at zpol Sn124 loose g050/g060 (0.95 vs 0.84); other points are within 5%. The discrepancy may come from statistical dependence between NN proton and the neutron-fallback (zpol sample is small, amplifying the effect); worth verifying in the RK comparison.
- **C full reco:** NEBULA-hit subset only, suffers the same acceptance bias.

6.3 6.3 zpol R and Δp_x^{rot} figures

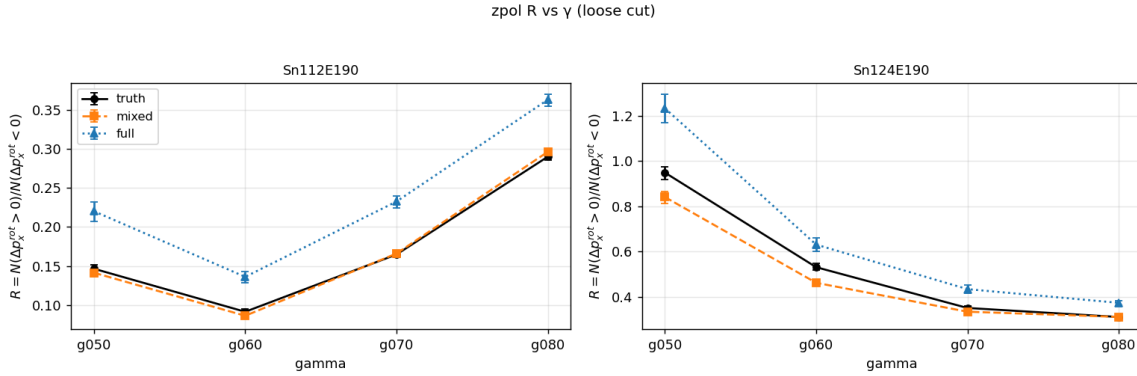


Figure 9: zpol R_x^{rot} vs gamma (loose cut).

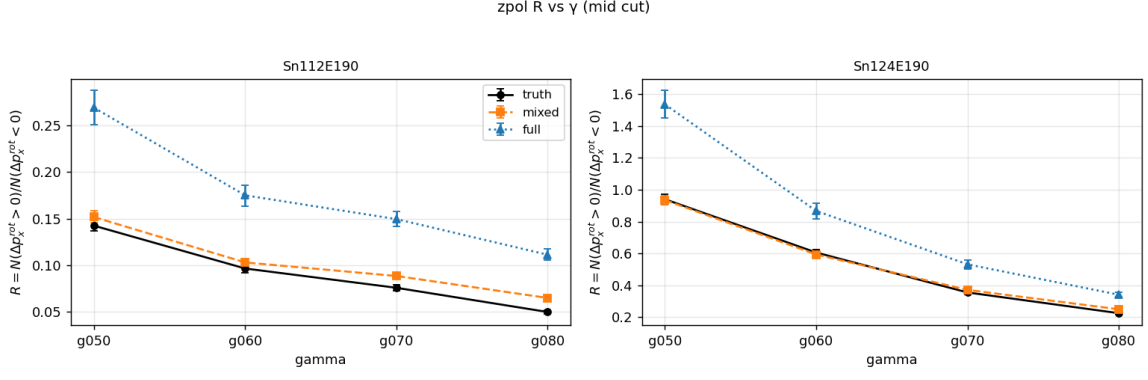


Figure 10: zpol R_x^{rot} vs gamma (mid cut).

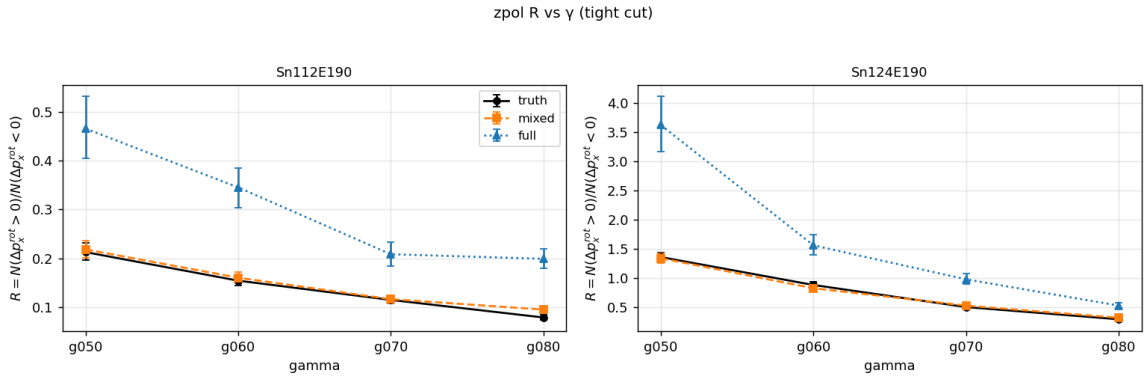


Figure 11: zpol R_x^{rot} vs gamma (tight cut).

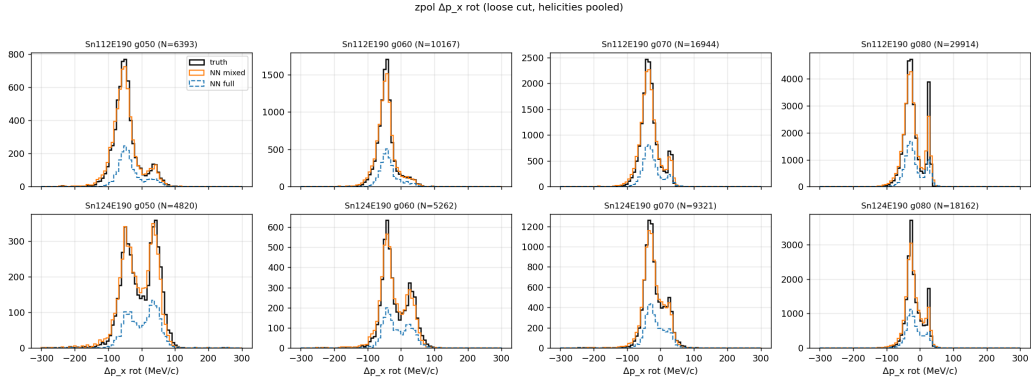


Figure 12: zpol Δp_x^{rot} distribution (loose cut).

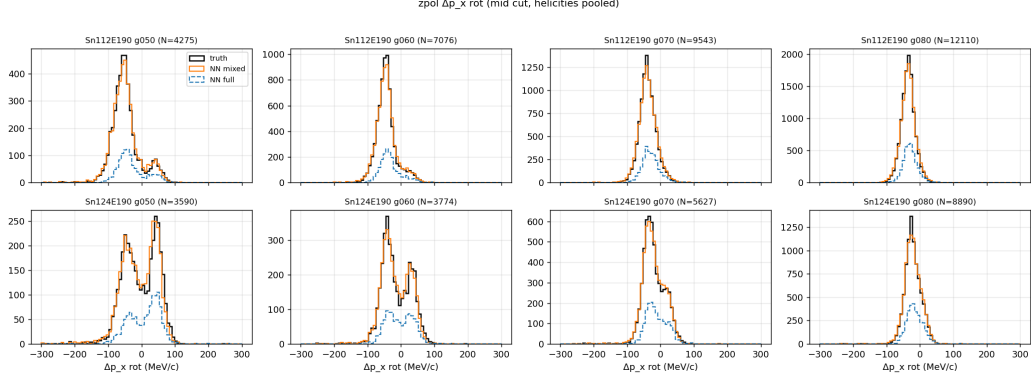


Figure 13: zpol Δp_x^{rot} distribution (mid cut).

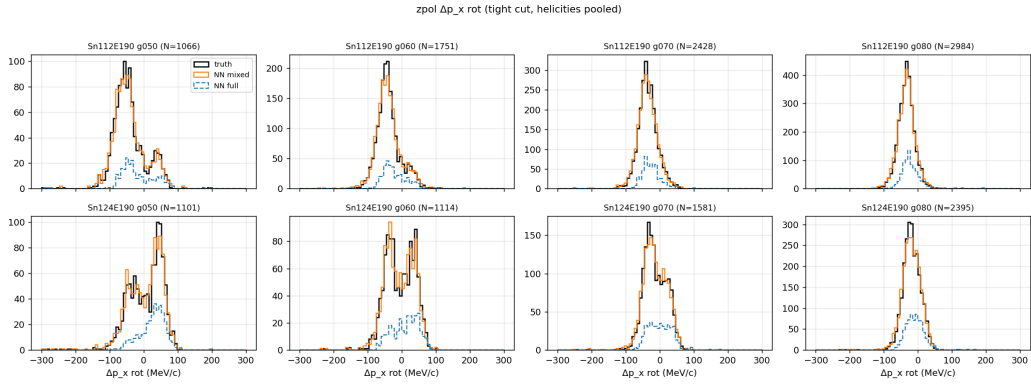


Figure 14: zpol Δp_x^{rot} distribution (tight cut).

6.4 6.4 zpol per-particle residuals

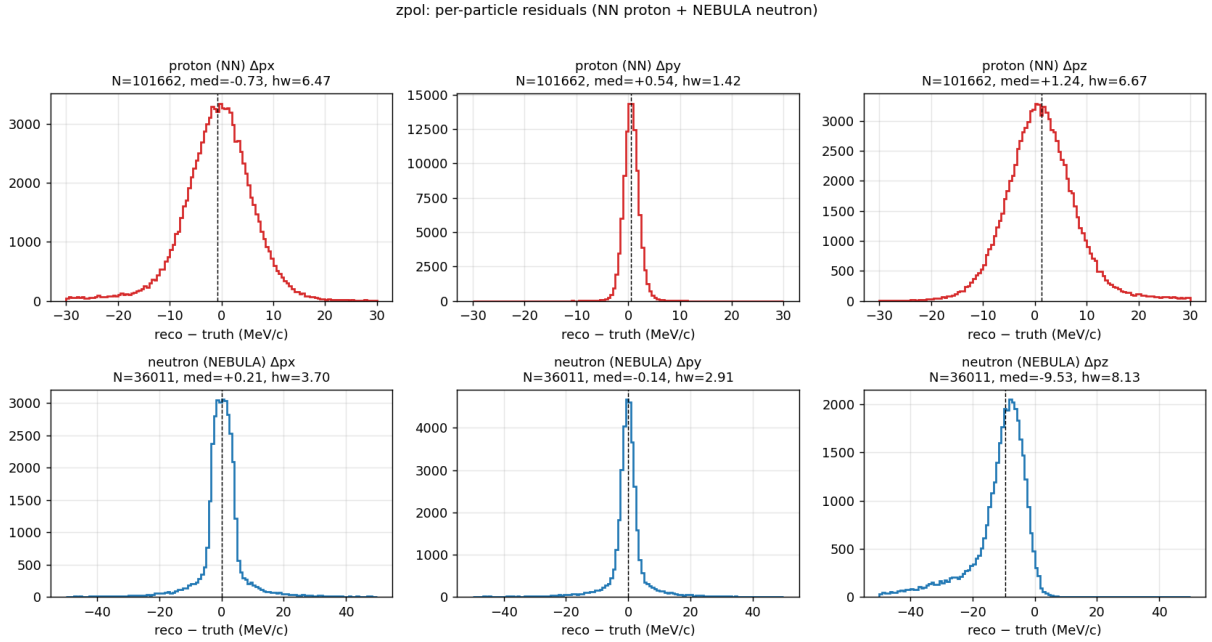


Figure 15: zpol per-particle residuals. Top: NN proton; bottom: NEBULA neutron.

zpol training-domain coverage is 91.3% (vs ypol 97.4%); residuals are slightly wider than ypol but at the same order of magnitude — not a meaningful perturbation to R.

6.5 6.5 zpol reco vs truth 2D

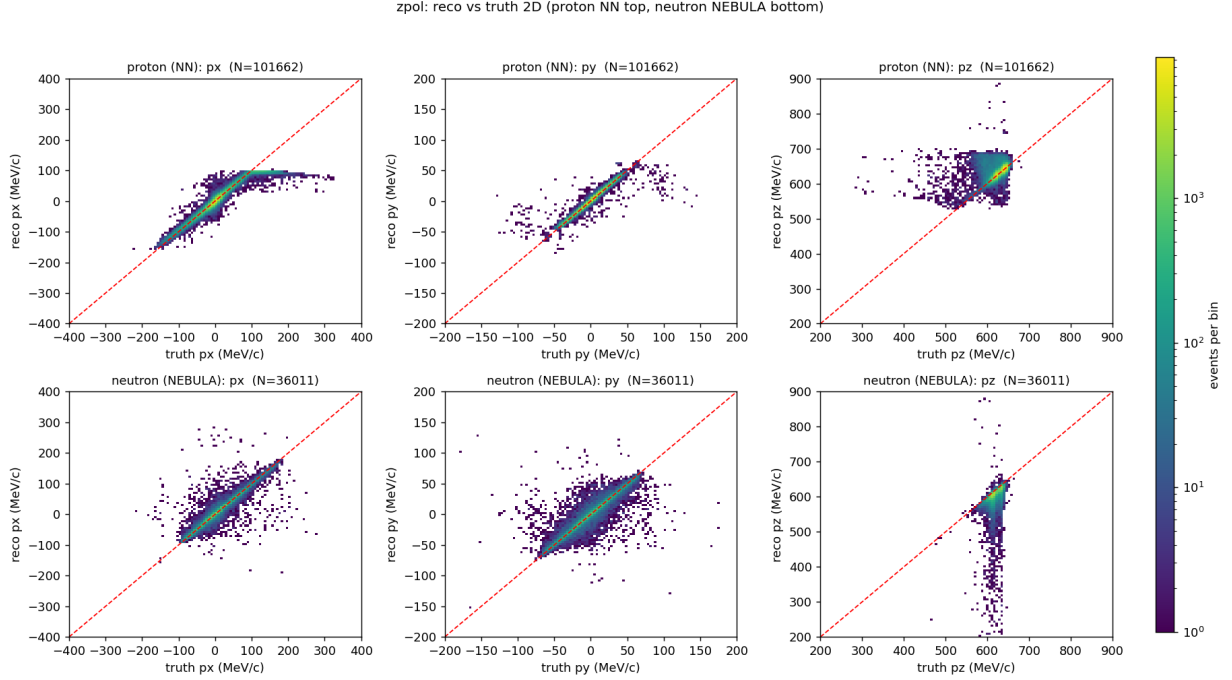


Figure 16: zpol reco vs truth 2D histograms. Top: NN proton; bottom: NEBULA neutron. Red dashed line $y = x$. All 6 panels share a common LogNorm color map; the right-side colorbar gives events per bin. Proton diagonal bends visibly outside the training domain ($|p_T| > 100$ MeV/c); neutron shows the same 12 cm NEBULA-bar X -discrete band structure as ypol.

7 7. Conclusion & next steps

1. The production NN model runs stably as forward inference on breakup g4output, producing proton output for 100% of cases.
2. ypol Sn124 R signal is clean and decreases monotonically with gamma under all three cuts; zpol Sn124 signal is even stronger, with larger R range.
3. NN B mixed vs A truth agrees to $< 3\%$ on ypol and $< 11\%$ on zpol; the latter is attributed to small statistics + neutron-fallback dependence and will be verified in the RK comparison.
4. NN proton residual σ on breakup matches elastic; training-domain coverage is 97% (ypol) / 91% (zpol), no significant extrapolation.

Follow-up reports:

- **Next:** same spana03 g4output, RK reconstruction (after running on spana03), as a standalone report `rk_breakup_phys_<date>_zh.tex`.
- **After that:** `nn_vs_rk_breakup_phys_<date>_zh.tex`, head-to-head, focusing on (a) whether NN is significantly worse than RK in the large- p_T extrapolation slice, and (b) whether the R deviation between the two methods is driven by a common NEBULA systematic.

Artifacts

- rsync script: `scripts/analysis/nn_breakup_phys/00_rsync_from_spana03.sh`

- NN reco driver: `scripts/analysis/nn_breakup_phys/01_run_nn_reco.sh`
- Extraction macro: `scripts/analysis/nn_breakup_phys/02_extract_observables.C`
- Extraction driver: `scripts/analysis/nn_breakup_phys/run_extract_all.sh`
- R / Δp_x analysis: `scripts/analysis/nn_breakup_phys/03_analyze_r_breakup.py`
- Plotting: `scripts/analysis/nn_breakup_phys/04_make_figures.py`
- NN reco ROOT output: `data/reconstruction/breakup_nn_20260503/{y_pol,z_pol}/`
- Intermediate CSV: `build/test_output/nn_breakup_phys_20260503/csv/{ypol,zpol}/`
- R table / pass rates / joined df: `build/test_output/nn_breakup_phys_20260503/summary/{ypol,zpol}/`